Jay Dukare

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Summary

Results-driven Machine Learning Engineer with hands-on experience building, interpreting, and deploying predictive models using XGBoost and scikit-learn. Proven track record of developing end-to-end ML pipelines, extracting actionable insights using SHAP explainability, and deploying live applications via Streamlit. Possesses a strong engineering foundation, bridging the gap between intelligent software architectures and edge-hardware integration.

Education

P.E.S. Modern College of Engineering — Pune, Maharashtra
Bachelor of Engineering in Electronics and Telecommunication; SGPA: 8.26 — *Aug 2023 – Present*

Government Polytechnic — Pune, Maharashtra
Diploma in Electronics and Telecommunication Engineering; 84.39% — *Jul 2020 – Jul 2023*

Technical Skills

Languages: Python, C++, C
Libraries & Frameworks: Pandas, NumPy, scikit-learn, XGBoost, Matplotlib
Core ML Concepts: Supervised Learning, Classification, Regression, Feature Engineering, Model Evaluation
Tools & Deployment: Git, GitHub, Jupyter Notebook, VS Code, Streamlit

Machine Learning & Engineering Projects

Customer Churn Prediction Engine | *XGBoost, SHAP, Streamlit, Optuna* | [GitHub] — 2026
- Engineered a high-performance classification model utilizing the Telco dataset to predict customer attrition, achieving an 87%+ F1 score.
- Executed end-to-end ML pipelines encompassing rigorous data cleaning, feature engineering, and automated hyperparameter tuning via Optuna.
- Integrated SHAP (SHapley Additive exPlanations) to enhance model transparency, identifying contract types as the primary churn driver.
- Deployed a live interactive web application using Streamlit to facilitate real-time single-customer and batch predictions.

Predictive Maintenance for Manufacturing | *scikit-learn, Pandas, Feature Engineering* | [GitHub] — 2026
- Developed a predictive ML model analyzing industrial sensor telemetry (temperature, rotational speed, vibration) to forecast equipment failure and estimate Remaining Useful Life (RUL).
- Applied classification and regression algorithms to mitigate factory downtime, demonstrating a practical intersection between AI and industrial automation.

Movie Recommender System | *Python, Pandas, Collaborative Filtering* | [GitHub] — 2025
- Designed a robust recommendation engine leveraging collaborative filtering and matrix factorization techniques to suggest content based on historical user behavior.
- Optimized data processing workflows using Pandas and NumPy to handle dense user-item interaction matrices efficiently.

Experience & Achievements

Head, IoT Hackathon — 2023 – 2024
Mpulse Xtronica, P.E.S. Modern College of Engineering — *Pune, Maharashtra*
- Spearheaded a team-level technical event, coordinating cross-functional teams to deliver a successful hardware and software hackathon.

Finalist, CircuitVista 2K25 — 2025
Hardware Design Competition — *Pune, Maharashtra*
- Recognized for technical excellence and innovative embedded systems integration among top competitors.

Industrial Automation Intern — Jul 2022 – Sep 2022
Pretech Automation Pvt. Ltd. — *Pune, Maharashtra*
- Assembled and wired Programmable Logic Controller (PLC) control panels, validating system functionality in real-world production environments.
- Acquired foundational expertise in hardware-software integration for industrial automation workflows.